Entanglement Sudden Death in Interacting Quantum Circuits

Samyak Jain
Andover High School

The theoretical aspects of entanglement dynamics in quantum circuits have been at the forefront of quantum information theory research. In particular, random unitary circuit models provide a simple platform for studying entanglement dynamics in generic quantum circuits. Here, we study quantum entanglement dynamics in a random Clifford circuit model, interspersed with single-qubit noise described by quantum channels. We study entanglement dynamics in the presence of noise and competing unitary evolution numerically using Qiskit. The simulation presents numerical evidence that the dissipation time in noisy Clifford circuits is the same as that of noisy circuits. Our work lays the foundation for developing a generalized mathematical proof for the critical time at which a quantum state becomes fully separable when subject to noisy unitary dynamics. If proven, such a theory can be applied to real-world quantum devices and algorithms, assisting with developing fault-tolerant code and hardware.

Introduction

Quantum entanglement serves as a crucial resource for quantum computation, communication and cryptography (Nielsen & Chuang, 2010). However, in physical quantum systems, noisy environments often lead to decoherence and dissipation, degrading entanglement over time. In fact, entangled quantum states subject to dissipation may completely disentangle in finite time. This phenomenon is known as 'entanglement sudden death' and has been well studied in systems with few degrees of freedom (Yu & Eberly, 2009). Understanding how entanglement evolves in noisy environments is vital for the development of cutting-edge quantum technologies.

In this paper, we analyze the impact of local dissipation on entanglement dynamics within quantum circuits, simulating practical noisy backgrounds. Specifically, we study the evolution of a many-qubit quantum state subject to entangling random 2-qubit unitary gates and local single-qubit dissipation. Prior research has established a lower bound for the critical dissipation time t_c after which the quantum state is completely disentangled when analyzing circuits composed of only single-qubit depolarizing channels (Gong & Ashida, 2024). Here, we relax the constraint that the quantum state is subject to only dissipation and allow it to entangle as the time evolves via 2-qubit unitary evolution. In particular, we investigate 2-qubit random Clifford circuit models interspersed with single-qubit depolarizing channels and random quantum channels.

We employ numerical simulations using Qiskit to model the evolution of logarithmic negativity, a measure of bipartite mixed-state quantum entanglement under different noise strength. By varying the probability p of applying random noise, we compare how entanglement dissipates in our experiment and that described in previous research. Our findings contribute to a broader understanding of entanglement dynamics in noisy quantum systems, with implications for quantum error correction and fault-tolerant quantum computing.

Building a Noisy Circuit

We begin by creating an L-qubit quantum circuit with depth $2L$, where L is an even natural number. We prepare an initial L-qubit GHZ state, defined by

$$|\mathrm{GHZ}\rangle = \frac{|0\rangle^{\otimes L} + |1\rangle^{\otimes L}}{\sqrt{2}}$$

such that our initial density matrix is

$$\rho = |\mathrm{GHZ}\rangle \langle \mathrm{GHZ}| \,. \tag{1}$$

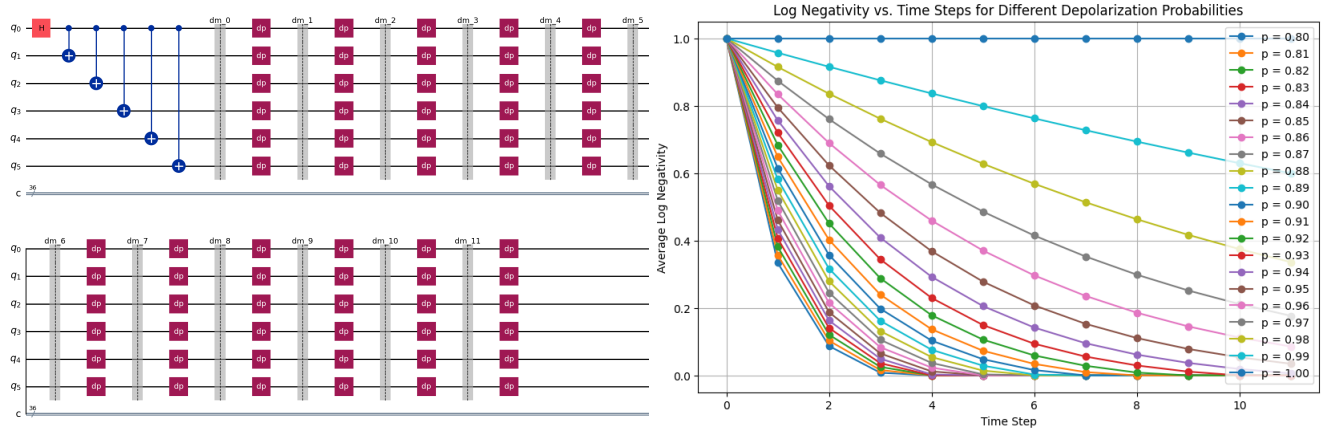

Figure 1. The left image is an example circuit consisting of only depolarizing channels, where $L = 6$ qubits. The right image is a plot of logarithmic negativity for different depolarization probabilities against time steps for the circuit shown on the left.

We then apply a 1-qubit depolarizing channel to each qubit in the range L at every time step, up to depth $2L$. This depolarizing channel is defined by

$$\mathcal{E}(\rho) = p\rho + (1-p)\frac{1}{2}I. \tag{2}$$

In other words, with probability p, the single-qubit depolarizing channel does nothing to the quantum state, and with probability $1-p$, we replace the qubit with a maximally-mixed state.

We have chosen to measure entanglement using logarithmic negativity because computing full separability of a quantum state is an NP-hard problem (Gharibian, 2008). Instead of determining the time until complete separability of a state, we measure the time it takes for the bipartite mixed-state entanglement to reach zero. Mixed-state entanglement is a less constrained measure of entanglement compared to complete separability, so this experiment does not exactly follow previous research but instead approaches the problem from a different perspective.

We aim to measure the logarithmic negativity of the density matrix at each time step in our circuit. Logarithmic negativity is defined as

$$E_N(\rho) = \log_2(2\mathcal{N} + 1) \tag{3}$$

where

$$\mathcal{N}(\rho) = \frac{\left\|\rho^{\Gamma_A}\right\|_1 - 1}{2} \tag{4}$$

is the quantum negativity. ρ^{Γ_A} denotes the partial transpose of ρ with respect to subsystem A and $\|\cdot\|_1$ denotes the trace norm. In this circuit, we present a bipartite system, such that subsystem A consists of $L/2$ qubits.

We then simulate this circuit using Qiskit's AerSimulator and scan logarithmic negativity over varying depolarizing probability p.

We show with numerical evidence that for an L-qubit circuit, the critical time where the bipartite mixed-state quantum entanglement becomes zero is governed by the equation

$$\lceil t_c \rceil = -\left\lceil \frac{\ln 3}{\ln p} \right\rceil \tag{5}$$

These results are consistent with the mathematical findings described in (Gong & Ashida, 2024). We now move on to creating a random Clifford circuit and incorporating this method of depolarization at each time step.

Clifford Gates

We aim to use Clifford gates in our random brickwork circuit. Clifford gates are defined as the elements of the Clifford group, which is the normalizer of the Pauli group within the group of all unitaries. If the Pauli group for an n-qubit system is defined as

$$\mathcal{P}_n = \{\pm 1, \pm i\} \times \{I, X, Y, Z\}^{\otimes n}$$

where I, X, Y, and Z are the Pauli matrices, then the Clifford group C_n is defined such that

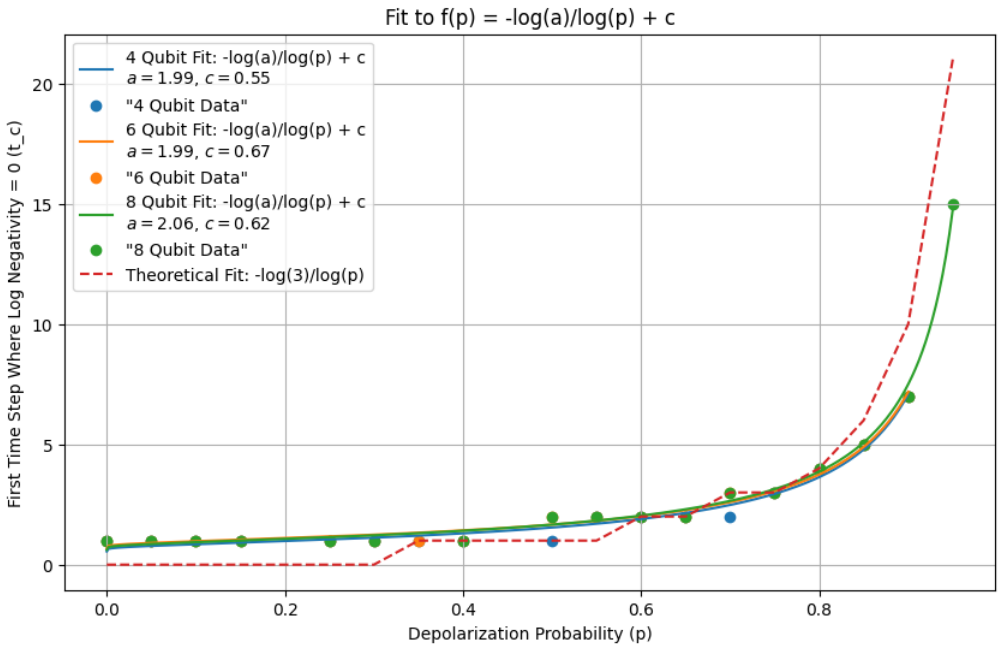

Figure 2. Best fit lines for circuits with $L = 4$ qubits, $L = 6$ qubits, and $L = 8$ qubits, with t_c plotted against depolarization probability p. The function $-\frac{\ln a}{\ln p} + c$. is optimized, where a and c are variables determined through SciPy's curve_fit module.

$$UPU^{\dagger} \in \mathcal{P}_n$$

for any U in C_n.

In other words, Clifford gates map Pauli strings to Pauli strings. Moreover, the Clifford group and therefore Clifford gates can be generated using only three generators: the H, S, and $CNOT$ gates.

We now begin constructing the brickwork Clifford circuit. To do this, we apply alternating 2-qubit Clifford unitaries. At each time step, qubits n and $n+1$ will be acted upon by a 2-qubit Clifford unitary, where n will cover all qubits in range L. This alternating pattern allows the system to become efficiently entangled as qubits interact with one another. This also allows for an evenly distributed spread of entanglement, since there are no localization effects caused by isolated pairs of qubits.

Incorporating Depolarizing Channels into the Clifford Circuit

We are now ready to incorporate the single-qubit noisy channels within our random Clifford circuit model. To do this, we begin by preparing an L-qubit GHZ state. Next, we apply alternating layers of depolarizing channels and Clifford gates, with even and odd two-qubit Clifford gates applied at every other time step.

We want to compare the effect of the random Clifford gates on the time until bipartite mixed-state entanglement reaches zero with that of the circuit with no Clifford gates. To do this, we can independently analyze the time steps using the density matrices and their respective logarithmic negativities. Figure 5 shows the number of time steps an 8-qubit circuit undergoes until the logarithmic negativity approaches zero. This is done using various probabilities p of depolarization. We observe that at high p values, the final state does not reach zero logarithmic negativity after $2L$ time steps, but at lower p values, the state does seem to reach zero logarithmic negativity when or before the circuit has evolved through $2L$ time steps.

Furthermore, Figure 5 shows that as the value of p decreases, the time until the logarithmic negativity reaches zero also decreases. This is consistent with the hypothesis and original t_c for a circuit composed only of depolarizing channels. Thus, we have numerically shown that t_c for a state subject to local dissipation is

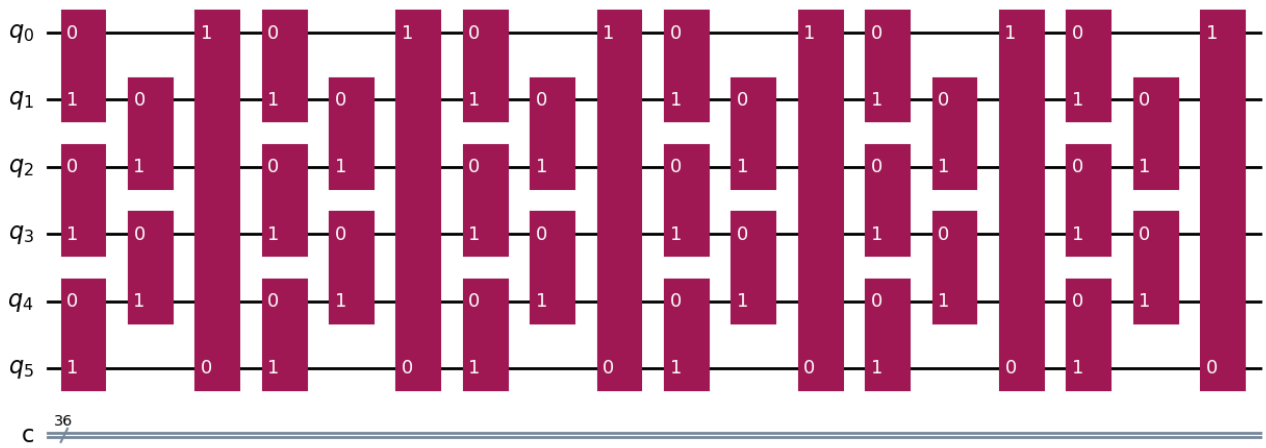

Figure 3. A random brickwork Clifford circuit generated when $L = 6$. The state is evolved through groups of three gates during every time step, such that it encompasses an even or odd cycle of alternating unitaries.

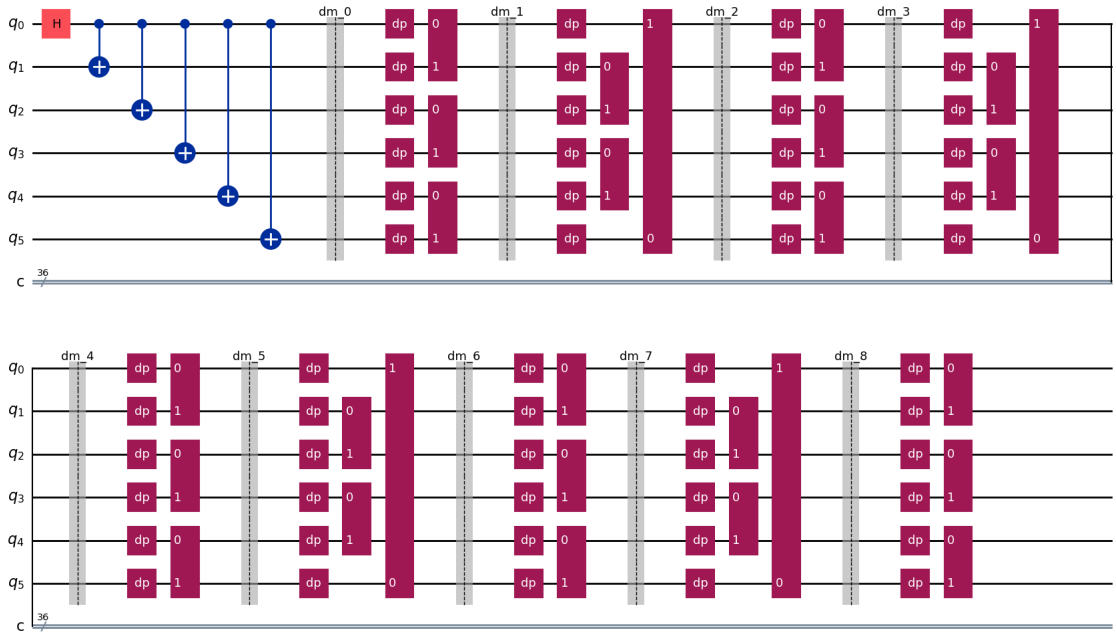

Figure 4. A Clifford circuit with $L = 6$ qubits and shorter depth for simplicity. This circuit is interspersed with single-qubit noisy channels.

the same as a state subject to local dissipation and random Clifford unitaries.

Random Quantum Channels

We now generalize our circuit to randomize within the set of all quantum channels instead of using only depolarizing channels. A random quantum channel (RQC) ϕ is defined as a completely positive trace preserving (CPTP) matrix. Completely positive means that

$$\phi \otimes I_n$$

is always positive, where I_n is the identity map on an n-dimensional Hilbert space, and trace preserving means that the trace of any density matrix is still equal to 1 after evolving it through $\phi(\rho)$.

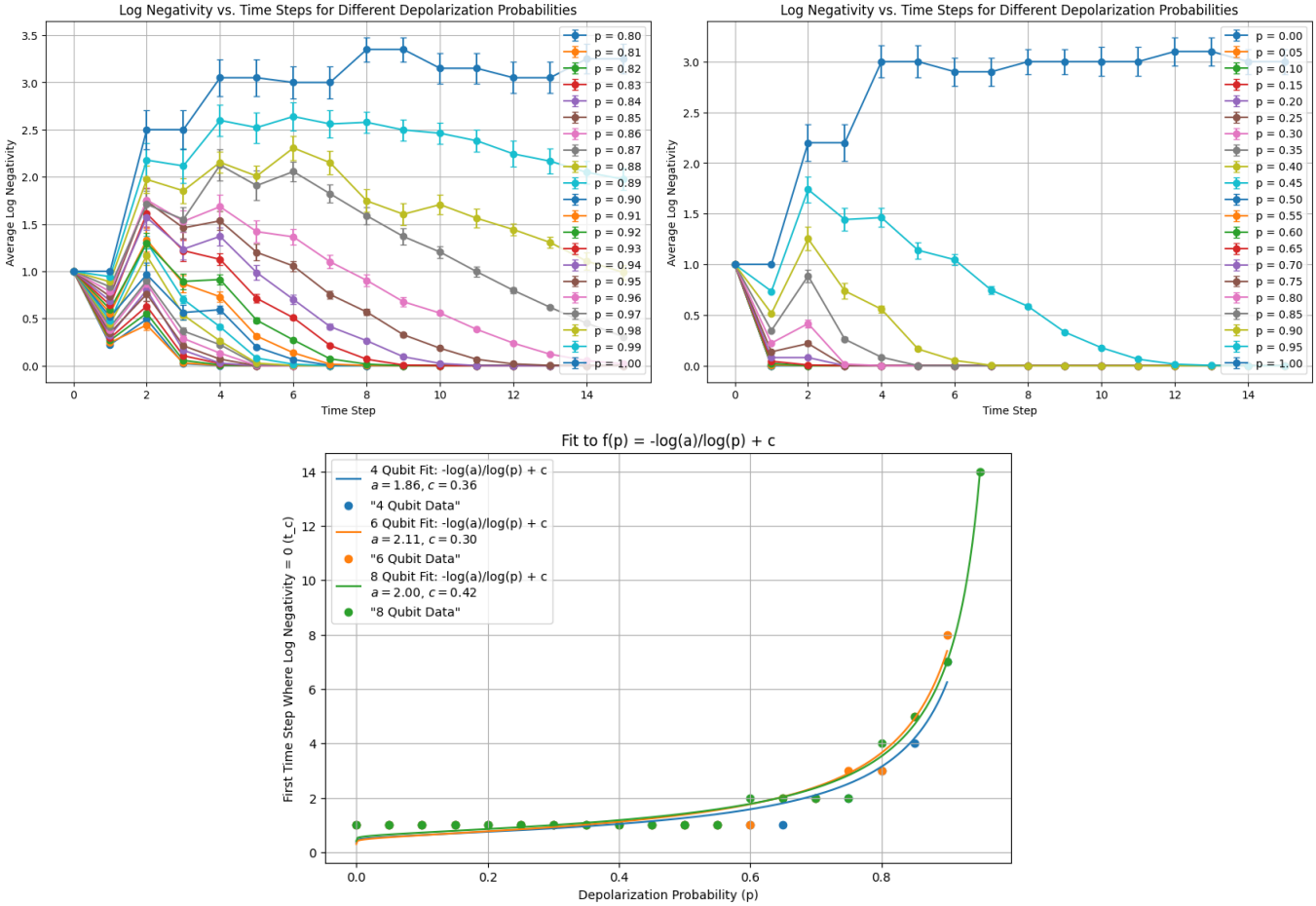

Figure 5. The left graph is a fine-tune scan over depolarizing probabilities between 0.8 and 1. The right graph is a scan over all depolarizing probabilities between 0 and 1. At lower p values, zero logarithmic negativity is reached at earlier time steps, whereas higher probabilities require more time steps to reach zero logarithmic negativity, if they do within depth $2L$. Both graphs also have error bars, which is calculated as $\frac{s}{\sqrt{n}}$, where s is the standard deviation and n is the number of runs. The bottom plot shows the lines of best fit for the first time step at which the logarithmic negativity of the state reaches zero at various L values. The plotted values of L are 4, 6, 8, and we see that they produce almost identical lines of best fit.

We first create a random Clifford circuit using the same procedure described in Section 3. We then apply an RQC before each 2-qubit Clifford gate with probability p. Instead of scanning over various probabilities of depolarization, we now scan over a range of probabilities that decide whether an RQC is applied to the state or not and subsequently analyze the logarithmic negativities of the state based on this probability.

Figure 7 shows a scan of RQC probabilities p from 0 to 0.1. All of the circuits were evolved for 50 time steps and averaged over 20 runs. The plotted points follow a similar trend as seen in the circuit composed of only depolarizing channels and the circuit composed of Clifford gates and depolarizing channels. The bounds on t_c seem to be the same for the three cases.

Conclusion

We have thus numerically proven that the time t_c until bipartite mixed-state quantum entanglement reaches zero for a state evolved through various circuit compositions that include depolarizing channels, Clifford unitaries, and random quantum channels follow similar patterns. The $-\left\lceil \frac{\ln 3}{\ln p} \right\rceil$ lower limit proven in (Gong

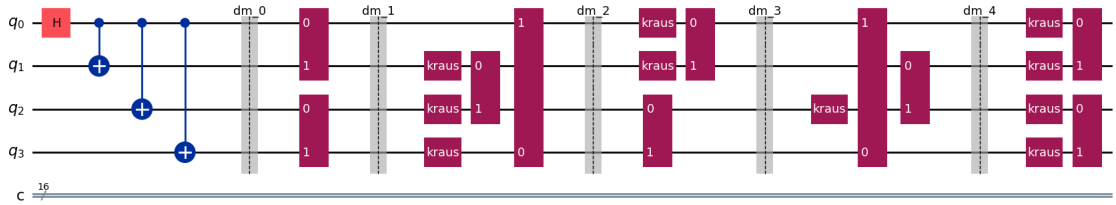

Figure 6. A visual depiction of a 4-qubit Clifford circuit with RQCs. The probability of an RQC being applied is $p = 0.5$ in this example, and if RQCs are applied, they are applied before the Clifford gates at each time step.

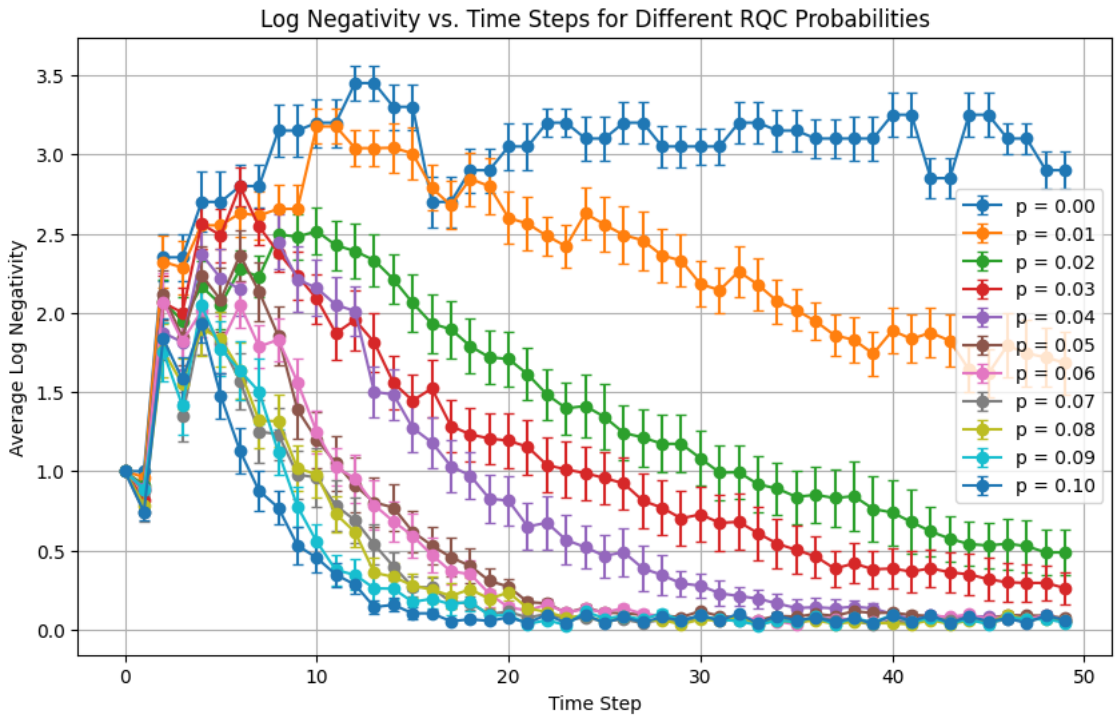

Figure 7. This plot shows time steps versus logarithmic negativity for an 8-qubit Clifford circuit that includes RQCs with probability p. Similar to the circuit interspersed with depolarizing channels, the graph produced shows that at higher p of RQC application, the state approaches zero logarithmic negativity more rapidly.

& Ashida, 2024) seem to be the same lower limit for the three circuit models explored within our simulation. We demonstrated this through Qiskit's density matrix circuit evolution method and SciPy's optimization features. These numerical results are evidence that entanglement in one-dimensional quantum states cannot survive local dissipation even when the state is simultaneously subject to entangling dynamics, such as 2-qubit unitaries. In this experiment, our source for local dissipation was depolarizing or random quantum channels, whereas our source of entanglement was Clifford gates.

The findings of this experiment lead to a wide range of possibilities for future work. This experiment's numerical nature calls for a mathematical generalization of the formulae derived in (Gong & Ashida, 2024) for circuits consisting only of depolarizing channels. Furthermore, more diverse sets of gates can also be explored. Clifford + T gates, matchgate circuits, diagonal gate sets, and low-degree polynomial circuits are other relevant possibilities that offer a wide range of applicability. Additionally, expanding this study to higher-dimensional systems, such as qutrits or qudits, could provide insight into entanglement dynamics in more complex quantum systems.

Determining entanglement dynamics in one-dimensional systems is extremely useful for quantum

error correction algorithms and quantum computing in general. Understanding the effects of local dissipation in degrading entanglement can help optimize error-correcting codes and improve the design of fault-tolerant quantum devices. We can assess the feasibility of the theoretical predictions described in this paper by extending such experiments to real quantum hardware, bridging the gap between quantum information theory and practical computing applications.

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